# Chapter 2 — Force and Motion

## Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

**1.** Dot product A·B equals: (a) AB tan θ (b) AB cos θ (c) AB sin θ (d) A/B

**2.** Cross product magnitude: (a) AB sin θ (b) A/B (c) AB tan θ (d) AB cos θ

**3.** Range is maximum at: (a) 45° (b) 60° (c) 90° (d) 30°

**4.** At top of projectile, vertical velocity: (a) zero (b) g (c) maximum (d) u

**5.** Impulse equals: (a) change in energy (b) change in momentum (c) F × d (d) m × a

**6.** In elastic collision, conserved: (a) momentum only (b) momentum and K.E. (c) neither (d) K.E. only

**7.** Airbags reduce force by: (a) larger Δp (b) no impulse (c) shorter time (d) longer impact time

**8.** If |A+B| = |A−B|, angle: (a) 180° (b) 45° (c) 90° (d) 0°

## Q.2 — SHORT QUESTIONS (6 × 2 = 12)

**1.** Why is the dot product of perpendicular vectors zero?

**2.** Why does the horizontal velocity of a projectile remain constant?

**3.** Why do complementary angles give the same range?

**4.** How does increasing impact time reduce the force?

**5.** Why is momentum conserved in a collision?

**6.** How does an elastic collision differ from an inelastic collision?

## Q.3 — LONG QUESTIONS (2 × 8 = 16)

**Q.3 (a)** Derive expressions for time of flight, maximum height and range of a projectile. **(4)**

**(b)** Show that the range is maximum at 45° and complementary angles give equal ranges. **(4)**

**Q.4 (a)** State the principle of conservation of momentum and apply it to an elastic collision in one dimension. **(4)**

**(b)** Explain impulse and its relation to momentum with everyday examples. **(4)**

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